

Fig. 6: An actual-size, single-side PCB for the RF receiver

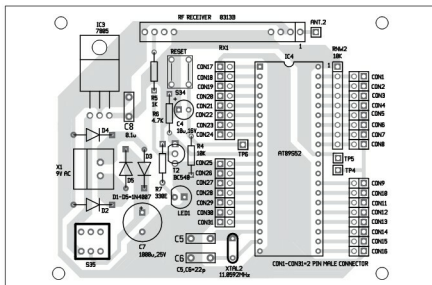


Fig. 7: Component layout for the PCB in Fig. 6

1's complement and '7' is the end of the frame. For example, if switch S1 is pressed, the data transmitted would be '9' '8' '8' 'A' 'inv of A' '7'.

Now consider reception of a code in a sequence like '9' '8' '8' 'A' 'inv of A' '7'.

'9' will be rejected since it is a junk

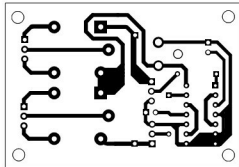


Fig. 8: Actual-size, single-side PCB for appliance driver

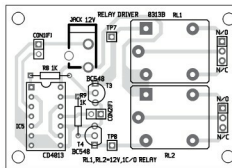


Fig. 9: Component layout for the PCB in Fig. 8

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

frame but Rx module will stabilise for next reception. ASCII code will be received after the reception of two consecutive '8'.

ASCII code is compared with its inverse to make sure that the correct code is received. According to conditional statements, port pin P0.0 of the receiver will go low and the corresponding appliance will turn on. Similarly, other appliances are controlled.

Construction

Actual-size, single-side PCB for the remote is shown in Fig. 4 and its component layout in Fig. 5. PCBs for the receiver and appliance driver are shown in Figs 6 and 8 and their component layouts in Figs 7 and 9, respectively. Assemble the circuits on PCBs to minimise assembly errors and to save time. Carefully assemble the components and double-check for any overlooked error. Use IC base for the microcontroller. Before inserting the IC, check the supply voltage. Suitable male connectors CON1 through CON31 are provided on the PCB of receiver for appliance driver connections.

The appliance driver PCB supports two relays. You can use multiple PCBs for more than two appliances. Connect CON1(F) and CON2(F) connectors of the appliance driver PCB to CON1(M) and CON2(M) connectors of the receiver PCB. Power-on 'the transmitter and receiver circuits. A 12V adaptor is used to power the appliance driver circuit. Now press switches S1 and S2 and check whether relays RL1 and RL2 energise. Appliances are connected through contacts of the relay. To verify correct functioning of the circuit, check the test points mentioned in the table. ●

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